Cairo University
Faculty of Engineering
Computer Engineering Department

Cairo University

Remote Sensing and Satellite Imagery

Dr. Sandra Wahid

Types of Image Data Errors

Radiometric Errors	Errors in the measured brightness values of the pixels.
Geometric Errors	Errors between the actual image acquisition (sensor) coordinate data and ideal image coordinate data.

<u>Note the following terminology:</u>

BV: Brightness Value *or* **DN**: Digital Number

Sources of Radiometric Distortion

- Two broad types of radiometric distortion:
  - The **distribution of brightness** over an image in a given band can be different from that in the ground scene.
  - The **relative brightness** of a single pixel from band to band can be distorted.

- Both types can result from the presence of the **atmosphere** as a transmission medium through which radiation must travel from its source to the sensors, and can also be the result of **instrumentation effects**.

Correcting Instrumentation Errors

- Common remote sensing radiometric errors caused by instrumentation effects:
  - Random bad pixels (shot noise)
  - Line drop-outs
  - Line striping

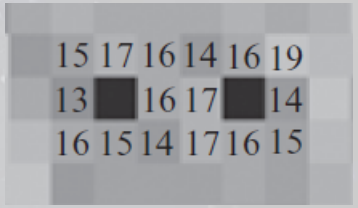
Random Bad Pixels (Shot Noise)

- Sometimes an individual detector does not record spectral data for an **individual pixel**.
  - When this occurs **randomly**, it is called a **bad pixel**.
  - When there are **numerous random** bad pixels found within the scene, it is called **shot noise** because it appears that the image was shot by a shotgun.
- Normally these bad pixels contain values of **0 or 255** (in 8-bit data) in one or more of the bands.
- Shot noise is identified and repaired using the following methodology:
  - First locate each bad pixel in the band k dataset: a simple **thresholding** algorithm makes a pass through the dataset and flags any pixel ($\boldsymbol{BV_{i,j,k}}$) having a brightness value of zero (while the surrounding pixels are not 0).
  - Once identified, it is then possible to evaluate the eight pixels surrounding the flagged pixel:

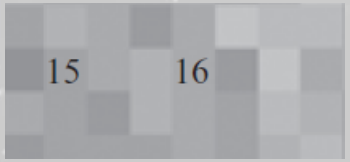
	col_{j-1}	col_j	col_{j+1}
row_{i-1}	BV_1	BV_2	BV_3
row_i	BV_8	BV_{ijk}	BV_4
row_{i+1}	BV_7	BV_6	BV_5

The mean of the eight surrounding pixels is computed and the value substitutes for $\boldsymbol{BV_{i,j,k}}$ in the corrected image:

$$BV_{i,j,k} = \text{Int}\left[\left(\sum_{i=1}^{8} BV_i\right)/8\right]$$



After Correction:

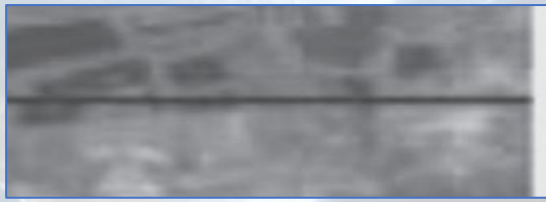


Line Drop-Outs

- If a detector in a linear array fails to function, this can result in an **entire line of data** with no spectral information.
- The bad line is commonly called a line drop-out and contains brightness values equal to zero.

- Methodology:
  - First locate each bad line in the image dataset: calculate the average ***BV*** per scan line for the entire scene. The average ***BV*** for each scan line is then compared with the scene average. Any scan line deviating from the average by more than a designated **threshold** value is identified as defective.
  - Once identified, it is then possible to collect the value for a pixel in the **preceding** line ($BV_{i-1,j,k}$) and the value for a pixel in the **succeeding** line ($BV_{i+1,j,k}$) and assign the output pixel ($BV_{i-1,j,k}$) in the drop-out line the **average** of these two brightness values:

$$BV_{i,j,k} = \mathrm{Int}\left(\frac{BV_{i-1,j,k} + BV_{i+1,j,k}}{2}\right)$$



89	89	96	94	98	108	111	110	94	80
104	108	93	92	97	94	86	81	75	83
0	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
131	143	107	101	122	127	88	86	87	90
122	155	147	115	145	155	101	91	92	95

89	89	96	94	98	108	111	110	94	80
104	108	93	92	97	94	86	81	75	83
118	**126**	**100**	**97**	**110**	**111**	**87**	**84**	**81**	**87**
131	143	107	101	122	127	88	86	87	90
122	155	147	115	145	155	101	91	92	95

Line Striping

- Sometimes a detector does not fail completely, but simply goes out of **radiometric adjustment**.
- For example, a detector might record spectral measurements that are almost uniformly 20 brightness values greater than the other detectors for the same band.
- The result would be an image with systematic, noticeable lines that are brighter than adjacent lines. This is referred to as ***n-line striping***. The correction is referred to as ***destriping.***
- The maladjusted line contains valuable information, but should be corrected to have approximately the same radiometric scale as the data collected by the properly calibrated detectors associated with the same band.
- To repair systematic n-line striping, it is first necessary to identify the miscalibrated scan lines in the scene.
- This is usually accomplished by computing a **histogram** of the values for each of the n detectors that collected data over the entire scene.

Line Striping

- If one detector's mean or median is significantly different from the others, it is probable that this detector is out of adjustment.
- Consequently, every line and pixel in the scene recorded by the maladjusted detector may require a bias (**additive or subtractive**) correction or a more severe gain (**multiplicative**) correction.
- This type of n-line striping correction
  - adjusts all the bad scan lines so that they have approximately the same radiometric scale as the correctly collected data.
  - improves the visual interpretability of the data (makes it look better).

Line Striping

- <u>Another Method:</u>
  - Correction of radiometric mismatches among the detectors can be carried out by adopting one detector as a **standard** and adjusting the brightness of all pixels recorded by each other detector so that their mean brightness and standard deviations match those of the standard detector.
  - That operation can be implemented by:

$$y = \frac{\sigma_d}{\sigma_i} x + m_d - \frac{\sigma_d}{\sigma_i} m_i$$

Why this accomplishes the goal??

- x is the original brightness for a pixel
- y is its new (destriped) value
- m_d and σ_d are the reference values of mean brightness and standard deviation for the standard detector.
- m_i and σ_i are the values of mean brightness and standard deviation for the detector under consideration.

- Sometimes an *independent* reference mean and standard deviation is used.

Correcting Atmospheric Errors

- When an atmosphere is present there are two effects that must be taken into account: ***scattering*** and ***absorption*** by the particles.

- These lead to:
  - Fine detail in image data is obscured: *haze* in the imagery
  - The presence of the *absorption lines* superimposed by water vapour and other atmospheric constituents.

- Approximate techniques are available for atmospheric correction that depend directly on the measurements of the recorded image data.

- These are important when detailed data on the atmosphere, and particularly the water vapour content, are not available.

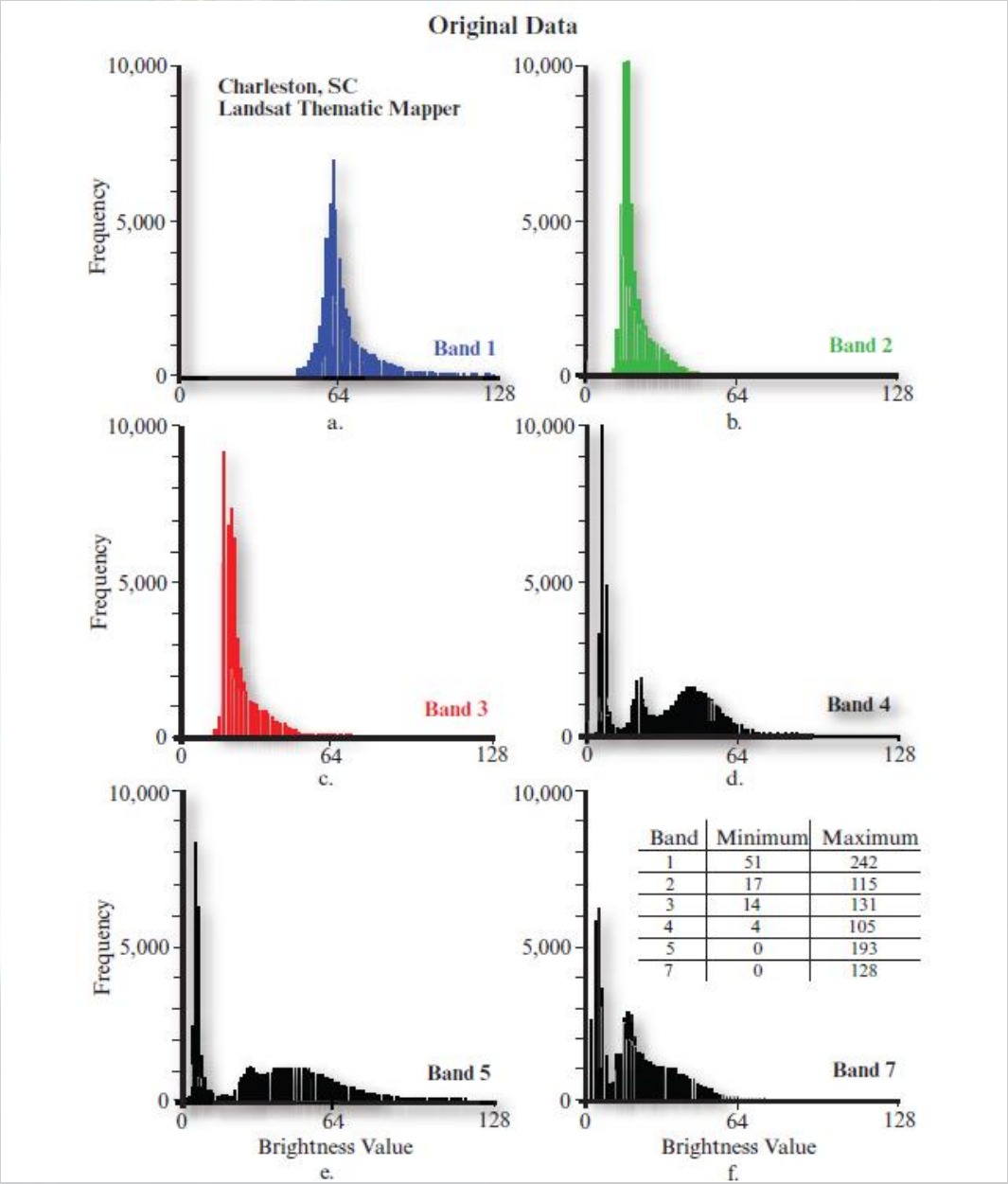
- The most common include:
  - Haze Removal by Dark Subtraction, also known as: Dark Object Subtraction

Haze Removal by Dark Subtraction

- It makes the assumption that each band of data for a given scene should contain some pixels at or close to **zero** brightness value but that atmospheric effects has added a **constant level** to each pixel in each band.

- Consequently, if histograms are taken of the bands (graphs of the number of pixels as a function of brightness value) the lowest significant occupied brightness value will be non-zero.

- The lowest occupied brightness value will be further from the origin for the *shorter wavelengths.*



Band	Minimum	Maximum
1	51	242
2	17	115
3	14	131
4	4	105
5	0	193
7	0	128

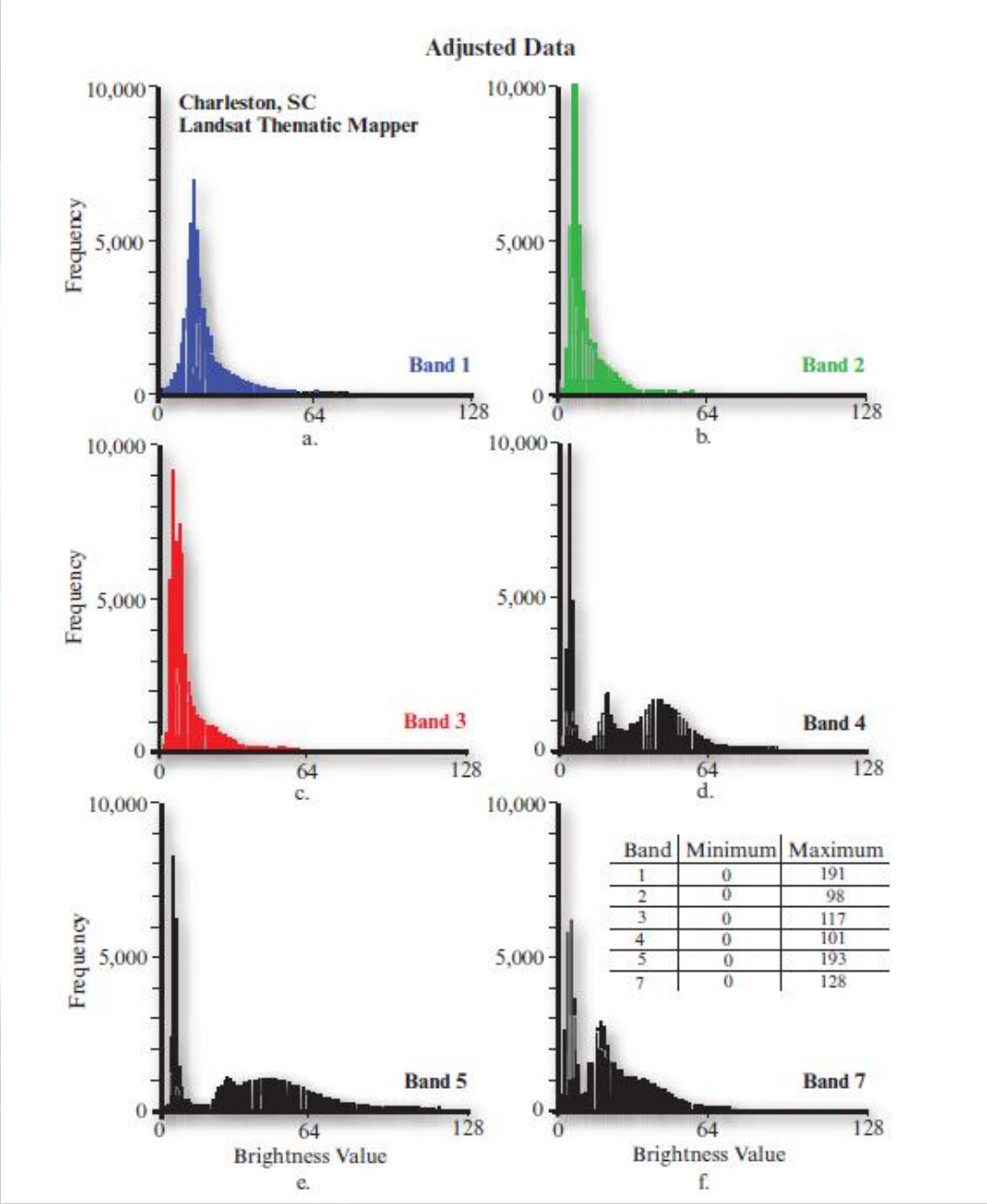
Haze Removal by Dark Subtraction

- If the histograms are shifted to the left so that zero values appear in the data, the effects of atmospheric scattering will be somewhat minimized.

- This simple algorithm is based on a **subtractive bias** established for each spectral band (the minimum *BV* in the histogram):

$$\text{output } BV_{i,j,k} = \text{input } BV_{i,j,k} - \text{bias}$$

- In the example the biases are chosen as follows:
  - Band1: bias=51
  - Band2: bias=17
  - Band3: bias=14
  - Band4: bias=4
  - Band5 and Band7: bias=0



Band	Minimum	Maximum
1	0	191
2	0	98
3	0	117
4	0	101
5	0	193
7	0	128

Sources of Geometric Distortion

- Sources of geometric distortion include:
  - The rotation of the earth during image acquisition.
  - Variations in platform altitude and velocity.
  - The wide field of view of some sensors.
  - The curvature of the earth.
  - Sensor non-idealities.

- There are two techniques that can be used to correct the various types of geometric distortion present in digital image data:

Model the nature and magnitude of the **sources of distortion** and use the model to establish correction formulas.

This approach is effective when the types of distortion are well characterized, such as that caused by earth rotation.

Establish **mathematical relationships between the addresses of pixels in an image and the corresponding coordinates of those points on the ground** (via a map)→ **Image-to-Map Registration/Rectification**

This approach is the **most commonly used** and is independent of the analyst's knowledge of the source and type of distortion.

Use of Mapping Functions for Image Correction

- An assumption made in this procedure is the availability of a map of the region covered by the image, which is correct geometrically.
- We then define two Cartesian coordinate systems:
  - one describes the location of points in the map ***(x,y)***
  - the other defines the location of pixels in the image ***(u,v) or (x',y')***
- Suppose that the two coordinate systems can be related via a pair of mapping functions:

$$u = f(x, y)$$
$$v = g(x, y)$$

- First, we define a grid over the map to act as the grid of pixel centers for the corrected image, referred to as the ***display grid***
- We then move over the display grid pixel center by pixel center and use the mapping functions above to find the pixel in the image corresponding to each display grid position.
- Those pixels are then placed on the display grid.
- At the conclusion of the process, we have a geometrically correct image built up on the display grid using the original image as a source of pixels.
- <u>Problems:</u>

1. We do not know the explicit form of the ***mapping functions***.
2. Even if we did, for a given display grid location they may not point exactly to a pixel in the image. In such a case some form of ***interpolation*** will be required.

Mapping Polynomials and the Use of Ground Control Points

- Since explicit forms for the mapping functions are not known

→they are usually approximated by **polynomials** (first, second or third degree).

Sometimes orders higher than three are used but care must be taken to avoid the introduction of errors worse than those to be corrected.

- In the case of second degree:

$$u = a_0 + a_1x + a_2y + a_3xy + a_4x^2 + a_5y^2$$

$$v = b_0 + b_1x + b_2y + b_3xy + b_4x^2 + b_5y^2$$

x/u (column) and *y/v* (row) coordinates

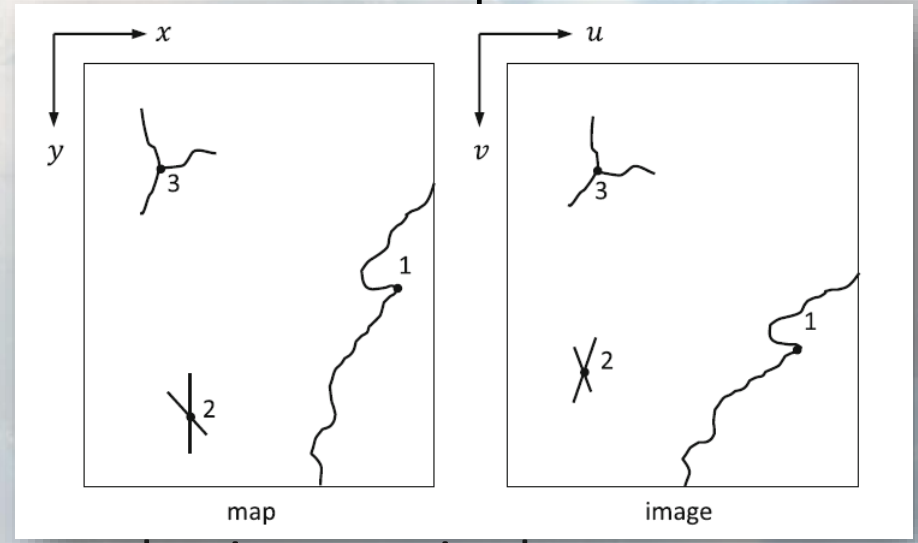
Known as: **Output-to-Input (Inverse) Mapping**
(u,v)→ original ***input*** image to be rectified
(x,y)→ ***output***-rectified image or map

- If the coefficients *ai* and *bi* were known then the mapping polynomials could be used to relate any point in the map to its corresponding point in the image.

- Coefficients values can be estimated by identifying sets of features on the map that can also be identified on the image. Those features, referred to as **ground control points (GCPs)** or just control points (CPs).

Mapping Polynomials and the Use of Ground Control Points

- These could be road intersections, street corners, airport runway intersections, sharp bends in rivers, coastline features, … etc.
- Enough are chosen, as pairs on the map and image so that the coefficients can be estimated by **substituting** the coordinates of the control points into the mapping polynomials to yield sets of equations in *ai* and *bi*.



- The minimum number of control points required:
  - for first order polynomial: 3
  - for second order polynomial: 6
  - for third order polynomial: 10
- In practice, significantly **more than those minimums** are chosen and the coefficients are evaluated using **least squares regression/estimation**.

→To avoid excessive influence on the estimated polynomial coefficients if any control points contain significant positional errors either on the map or in the image.

Problem 1 Solved

Ground Control Points

- Most image-to-map rectification still relies heavily on **human selection** of GCPs.
- A general rule is that there should be a distribution of control points around the **edges** of the image to be corrected, with a **scattering** of points over the body of the image →that is necessary to ensure that the mapping polynomials are well-behaved over the scene.
- After computing equations' coefficients, it is important to determine how well the coefficients account for the geometric distortion in the input image.

The method often used is root-mean-square error (**RMS_{error}**)

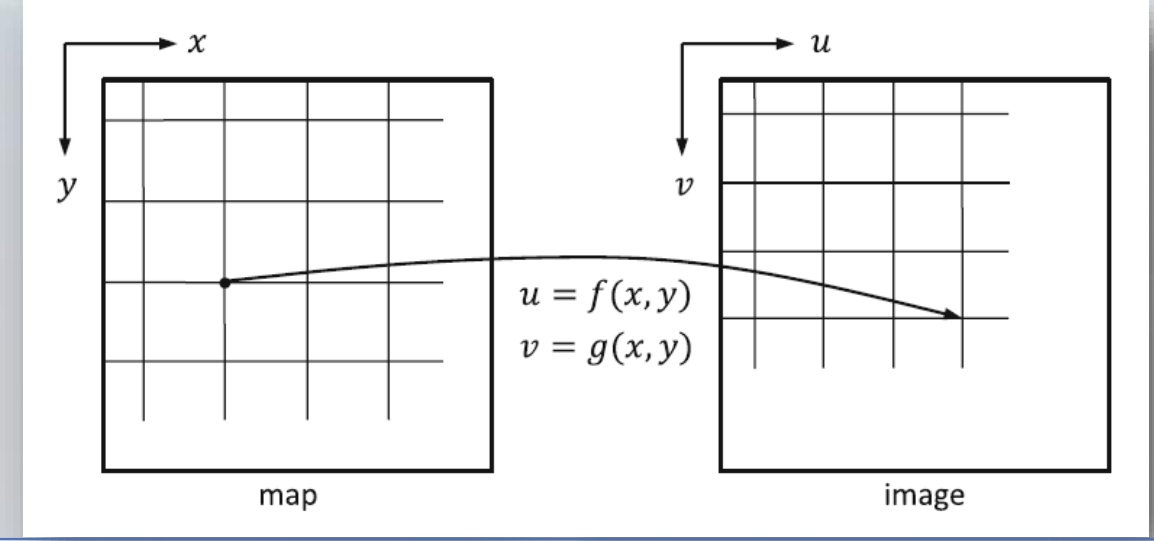
- Let x_{orig} and y_{orig} be the column and row coordinates of GCP in the original input image.
- Let x and y be the column and row coordinates of GCP in the reference map.
- Let x' and y' be the column and row coordinates of GCP from the equation.

→ Ideally $x' = x_{orig}$ and $y' = y_{orig}$

- For each GCP, compute: $$RMS_{error} = \sqrt{(x' - x_{orig})^2 + (y' - y_{orig})^2}$$
- By computing RMS_{error} for all GCPs, it is possible to:
  - 1) see which GCPs exhibit the greatest error 2) sum all the RMS_{error}
- The user specifies a threshold of acceptable total RMS_{error}. If the threshold is exceeded:
  - 1) delete the GCP that has the greatest RMS_{error} 2) recompute the coefficients 3) recompute the RMS_{error} for all points
  - This process continues until the total RMS_{error} is less than the threshold specified or too few points remain to perform a least-squares regression to compute the coefficients.

Building a Geometrically Correct Image

- Referred to as: **rectified image.**
- Having specified the mapping polynomials completely by the use of ground control points →the next step is to find points in the image that correspond to each location in the display grid.
- The spacing of that grid is chosen according to the **pixel size required** in the corrected image and need not be the same as that of the original, geometrically distorted version.
- For the moment suppose that the points located in the image correspond exactly to image pixel centers, even though that rarely happens in practice.
- Then those pixels are **simply transferred** to the appropriate locations on the display grid to build up the rectified image.



Using mapping functions to locate points in the image corresponding to particular display grid (map) positions

Resampling and the Need for Interpolation

- As expected, grid centers from the map-defined display grid will not usually project to exact pixel center locations in the image.

- Some decision now has to be made about what pixel brightness value should be chosen for placement on the new grid.

- Three principal techniques are used for this purpose:

Nearest Neighbor Resampling

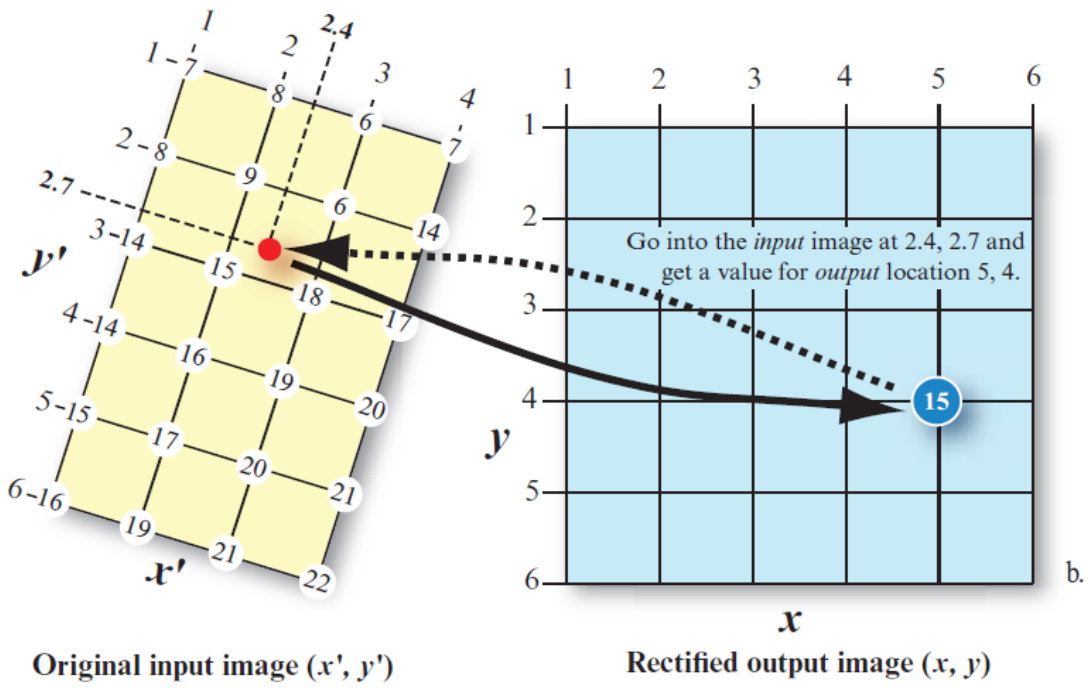
Bilinear Interpolation

Cubic Convolution Interpolation

Problem 2 Solved

Nearest Neighbor Resampling

- Simply chooses the actual pixel that has **its center nearest to the point located in the image**.
- That pixel value is transferred to the corresponding display grid location.
- This is the preferred technique if the new image is to be classified (e.g.: distinguishing one type of vegetation from another) because it consists only of original pixel brightness values, simply rearranged in position to give the correct image geometry.
- The method is only acceptable when the new and old pixel sizes and spacings are not too different.



$$x' = a_0 + a_1 x + a_2 y$$
$$y' = b_0 + b_1 x + b_2 y$$

Sample Point Location (column, row)	Value at Sample Point, Z	Distance from x', y' to the Sample Point, D
2, 2	9	$D = \sqrt{(2.4-2)^2 + (2.7-2)^2} = 0.806$
3, 2	6	$D = \sqrt{(2.4-3)^2 + (2.7-2)^2} = 0.921$
2, 3	15	$D = \sqrt{(2.4-2)^2 + (2.7-3)^2} = 0.500$
3, 3	18	$D = \sqrt{(2.4-3)^2 + (2.7-3)^2} = 0.670$

Bilinear Interpolation

- First-order or bilinear interpolation assigns output pixel values by interpolating brightness values in two orthogonal directions in the input image.
- It basically fits a plane to the **four pixel values** nearest to the desired position *(x',y')* in the input image and then computes a new brightness value based on the weighted distances to these points.
- The closer a pixel is to the desired location, the more weight it will have in the final computation of the average.
- The weighted average of the new brightness value (***BV***$_{wt}$) is computed using the equation:

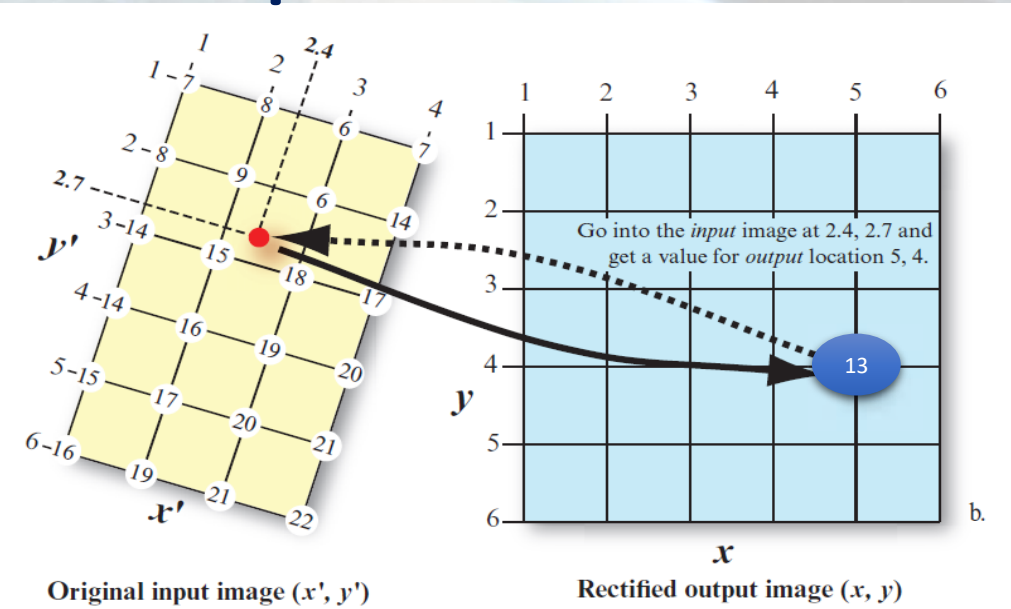
$$\text{Bilinear}_{BV_{wt}} = \frac{\sum_{k=1}^{4} \frac{Z_k}{D_k^2}}{\sum_{k=1}^{4} \frac{1}{D_k^2}},$$

Where:

- Z_k are the surrounding 4 data point values.
- D_k^2 are the distances squared from the point in question (x',y') to these data points.

Bilinear Interpolation

- Example:



Sample Point Location (column, row)	Value at Sample Point, Z	Distance from x', y' to the Sample Point, D	D_k^2	$\frac{Z}{D_k^2}$	$\frac{1}{D_k^2}$
2, 2	9	$D = \sqrt{(2.4-2)^2 + (2.7-2)^2} = 0.806$	0.65	13.85	1.539
3, 2	6	$D = \sqrt{(2.4-3)^2 + (2.7-2)^2} = 0.921$	0.85	7.06	1.176
2, 3	15	$D = \sqrt{(2.4-2)^2 + (2.7-3)^2} = 0.500$	0.25	60.00	4.000
3, 3	18	$D = \sqrt{(2.4-3)^2 + (2.7-3)^2} = 0.670$	0.45	40.00	2.222
				Σ 120.91	Σ 8.937
				BV_{wt} = 120.91/8.937 =13.53	

Truncate to 13

Cubic Convolution Interpolation

- Assigns values to output pixels in the same manner as bilinear interpolation, except that **sixteen input pixels** surrounding the location of the desired pixel are used.
- The weighted average of the new brightness value (***BV_{wt}***) is computed using the equation:

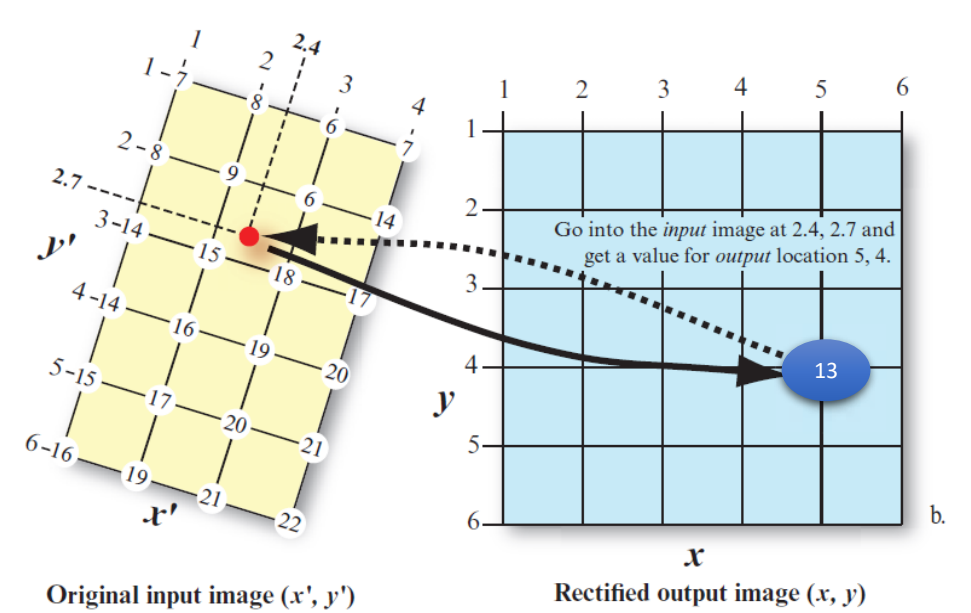
$$\text{Cubic Convolution}_{BV_{wt}} = \frac{\sum_{k=1}^{16} \frac{Z_k}{D_k^2}}{\sum_{k=1}^{16} \frac{1}{D_k^2}},$$

Where:

- Z_k are the surrounding 16 data point values.
- D_k^2 are the distances squared from the point in question (x',y') to these data points.

Cubic Convolution Interpolation

- Example:



$$x' = a_0 + a_1 x + a_2 y$$
$$y' = b_0 + b_1 x + b_2 y$$

Sample Point Location (column, row)	Value at Sample Point, Z	Distance from x', y' to the Sample Point, D	D_k^2	$\frac{Z}{D_k^2}$	$\frac{1}{D_k^2}$
1, 1	7	$D = \sqrt{(2.4-1)^2 + (2.7-1)^2} = 2.202$	4.85	1.443	0.206
2, 1	8	$D = \sqrt{(2.4-2)^2 + (2.7-1)^2} = 1.746$	3.05	2.623	0.328
3, 1	6	$D = \sqrt{(2.4-3)^2 + (2.7-1)^2} = 1.80$	3.24	1.852	0.309
4, 1	7	$D = \sqrt{(2.4-4)^2 + (2.7-1)^2} = 2.335$	5.45	1.284	0.183
1, 2	8	$D = \sqrt{(2.4-1)^2 + (2.7-2)^2} = 1.565$	2.45	3.265	0.408
2, 2	9	$D = \sqrt{(2.4-2)^2 + (2.7-2)^2} = 0.806$	0.65	13.85	1.539
3, 2	6	$D = \sqrt{(2.4-3)^2 + (2.7-2)^2} = 0.921$	0.85	7.06	1.176
4, 2	14	$D = \sqrt{(2.4-4)^2 + (2.7-2)^2} = 1.746$	3.05	4.59	0.328
1, 3	14	$D = \sqrt{(2.4-1)^2 + (2.7-3)^2} = 1.432$	2.05	6.829	0.488
2, 3	15	$D = \sqrt{(2.4-2)^2 + (2.7-3)^2} = 0.500$	0.25	60.00	4.000
3, 3	18	$D = \sqrt{(2.4-3)^2 + (2.7-3)^2} = 0.670$	0.45	40.00	2.222
4, 3	17	$D = \sqrt{(2.4-4)^2 + (2.7-3)^2} = 1.63$	2.65	6.415	0.377
1, 4	14	$D = \sqrt{(2.4-1)^2 + (2.7-4)^2} = 1.911$	3.65	3.836	0.274
2, 4	16	$D = \sqrt{(2.4-2)^2 + (2.7-4)^2} = 1.360$	1.85	8.649	0.541
3, 4	19	$D = \sqrt{(2.4-3)^2 + (2.7-4)^2} = 1.432$	2.05	9.268	0.488
4, 4	20	$D = \sqrt{(2.4-4)^2 + (2.7-4)^2} = 2.062$	4.25	4.706	0.235
				Σ 175.67	Σ 13.102
				BV_{wt} = 175.67 / 13.102 BV_{wt} = **13.41**	

Truncate to 13

Image-to-Image Registration

- Many applications in remote sensing (e.g. change detection) require two or more scenes of the same geographical region, acquired at different times, to be processed together.

- Two images can be registered to each other by registering each to a map coordinate base separately in the manner previously demonstrated.

- Alternatively, and particularly if georeferencing is not important, one image can be chosen as a **master, or reference**, to which the other, known as the **slave**, is registered.

- Same techniques as image-to-map registration are used.
- However, the coordinates are now the pixel coordinates in the master image rather than the map coordinates.

- A benefit in image-to-image registration is that only one registration step is required, by comparison to two if both are taken back to a map base.

Other Image Geometry Operations

- Intentional changes to image geometry can be performed including:
  - Image Rotation
  - Scale Changing and Zooming

Image Rotation:

To rotate an image in an anticlockwise sense by any specified angle ζ:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\zeta & \sin\zeta \\ -\sin\zeta & \cos\zeta \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

Negative the angle

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \cos\zeta & -\sin\zeta \\ \sin\zeta & \cos\zeta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Other Image Geometry Operations

Image Scale Changing and Zooming:

- The scales of an image in both the vertical and horizontal directions can be altered by the transformation:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

a and *b* are the desired scaling factors

- To resample the scaled image onto the display grid we use the inverse operation to locate pixel positions in the original image corresponding to each display grid position:

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 1/a & 0 \\ 0 & 1/b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- Interpolation is used to establish the actual pixel brightness values to use, since *u* and *v* will not normally fall on exact pixel locations.
- Frequently *a=b* so that the image is simply magnified called **zooming**.
- If the *nearest neighbor interpolation* procedure is used in the resampling process the zoom implemented is said to occur by ***pixel replication*** and the image will look progressively *blocky* for larger zoom factors.
- If *cubic convolution interpolation* is used there will be a *change in magnification* but the image will *not take on the blocky* appearance. Often this process is called ***interpolative zoom***.

Thank You